Shear Thickening Fluid Application in Protective Footwear System For Soccer Athletes

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Introduction & Problem Statement

Soccer athletes face frequent upper-foot injuries in the **toebox region** due to stud impacts during tackles and ball strikes. Traditional cleats prioritize ball control, leaving this area vulnerable to high localized forces. Previously added protection has relied on thicker materials or rigid TPU shells, which reduce comfort and increase weight.

- Common injuries: metatarsal fractures, bruising, chronic stress injuries.
- Traditional cleats prioritize ball feel, toe-box is unprotected.
- High-localized forces during gameplay cause significant injury risk.

Solution & Design

Hybrid Sandwich Composite

Outer Layer
Tough, flexible outer layer to protect the honeycomb, hold its shape, and ensure the STF does not leak from the product.

Core: 2, 3, and 5 mm Nomex Honeycomb:
Aramid fiber structure provides the primary geometric framework for energy absorption and impact resistance.

Active Element: Shear-Thickening Fluid (STF):
Non-Newtonian fluid filling the honeycomb cells, providing adaptive hardening upon sudden impact. Due to the initial shear thinning stage, and the general nature of STFs, the fluid will not thicken under the normal conditions. This will allow for full flexibility while playing, only thickening under impact.

Interface: High-Strength Adhesive
Urethane glue is spread evenly across TPU film and applied to the honeycomb.

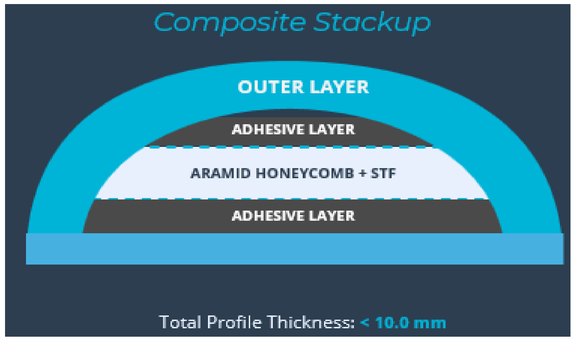

Figure 2: Breakdown of composite layers within the system

Design Objectives & Constraints

Objectives:

- Energy Dissipation: Reduce peak impact force transmitted to the foot.
- Stress Reduction: Distribute localized loads across a larger surface area.
- Performance: Maintain flexibility and natural foot movement.

Key Constraints:

- FIFA Compliance: No danger to other players.
- Mass: Total added weight of combined materials should not be overly heavy to maintain flexibility and comfort.
- Thickness: Thickness of total product should not impede performance
- Sustainability: Environmentally friendly materials.
- Social Equity: Accessible and affordable globally.

Materials Science: Shear Thickening Fluid (STF)

STFs are **Non-Newtonian fluids** whose viscosity increases with shear rate ("dilatant" behavior). Formulation: Fumed Silica in PEG-200.

Mechanism: Under impact shear, particles "hydrocluster," creating a temporarily solid-like state.

DST vs CST: Discontinuous Shear Thickening (DST) provides near-instantaneous hardening.

Model: Galindo-Rosales power-law viscosity function (n > 1).

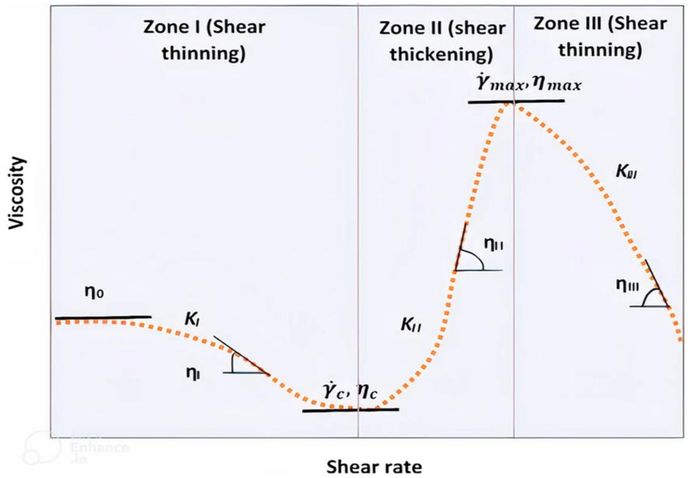

*Figure 3: DST modeling using the Galindo-Rosales method. Zarei, M.; Aalaie, J. Application of Shear Thickening Fluids in Material Development. J. Mater. Res. Technol.**2020**, 9, 10411–10433, doi: 10.1016/j.jmrt.2020.07.049*

Technical Approach/Experimental Setup

A custom **Drop Weight Test Rig** was fabricated using flanged bearings and steel poles to simulate high-velocity tackle impacts. The physical drop weight test rig design that would be used to attach the cleat studs and shoe toe box. The weight holding platform and physical weight will be added to the final design.

- Lubricated guiding poles for precise drop.
- High-speed video for rebound height analysis.

Figure 1: Test Rig outfitted with weight drop

Results

Compression and impact tests were conducted to evaluate the performance of the honeycomb-STF protective system in comparison to baseline materials.

Given the difference in the indentation values as well as the reduction in peak force, this indicates that the honeycomb–STF system effectively dissipates impact energy. This behavior aligns with the expected plateau region of honeycomb structures, where progressive cell collapse limits force transmission.

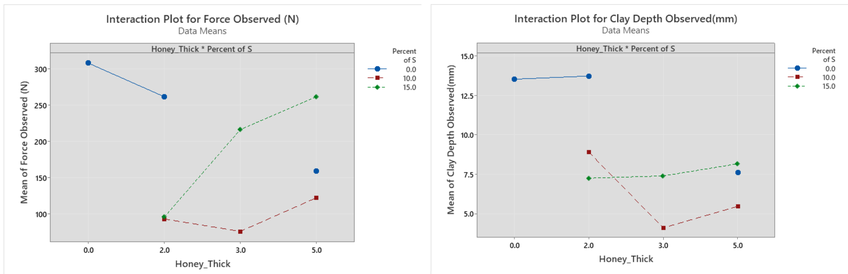

Figure 4: Mean force observed decreases with increased protection, the lowest force occurs at 3.0 mm thickness and 10% STF, indicating improved impact resistance

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References

Zarei, M.; Aalaie, J. Application of Shear Thickening Fluids in Material Development. *J. Mater. Res. Technol.*2020, *9*, 10411–10433, doi: 10.1016/j.jmrt.2020.07.049

Galindo-Rosales, F. J., Rubio-Hernández, F. J., Sevilla, A., & Ewoldt, R. H. (2011). How dr. Malcom M. Cross may have tackled the development of "an apparent viscosity function for shear thickening fluids." *Journal of Non-Newtonian Fluid Mechanics*, *166*(23–24), 1421–1424. https://doi.org/10.1016/j.jnnfm.2011.08.008